



Smarter Nutrition Through Data

A Machine-Learning Approach to Classifying Food Products by Nutri-Score

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Why This Project Matters

Nutri-Score helps simplify nutrition decisions

- Converts detailed nutrition tables into a single grade (A-E).
- Helps consumers compare products.

But many products don't display Nutri-Score

- Not mandatory in most countries.
- Users often rely on apps or guesswork when the label is missing.

Opportunity for machine learning

- Nutritional values per 100g are standardized and widely available.
- ML models can learn these patterns and predict the grade automatically.

Why it matters in practice

- Apps can display Nutri-Score for unlabeled products.
- Researchers can analyze food quality at scale.
- Consumers gain clearer guidance and healthier choices.

Research Question

Can machine learning accurately predict a food's
Nutri-Score using only its nutritional values?



Goal: Compare multiple models.
Identify best-performing approach.
Understand which nutrients matter most.

Data Summary

- **Name:** OpenFoodFacts
- **Data File:** Excel Spreadsheet (2D Array)
- **Structure:** Tabular
- **Rows:** 356,027
- **Features:** 163
- **Target:** numeric version of the Nutri-Score label (A-E $\rightarrow$ 1-5)



Data Cleaning

**Ensured
dataset quality
before any
modeling**

**Removed
incomplete or
unusable
entries**

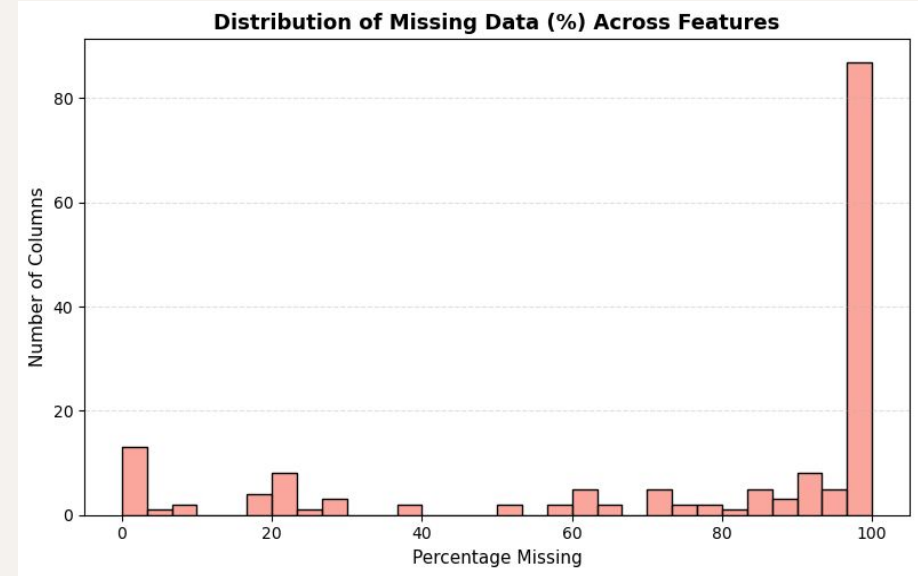
**Cleaned numeric
values and
standardized
units**

**Detected and
handled
outliers**

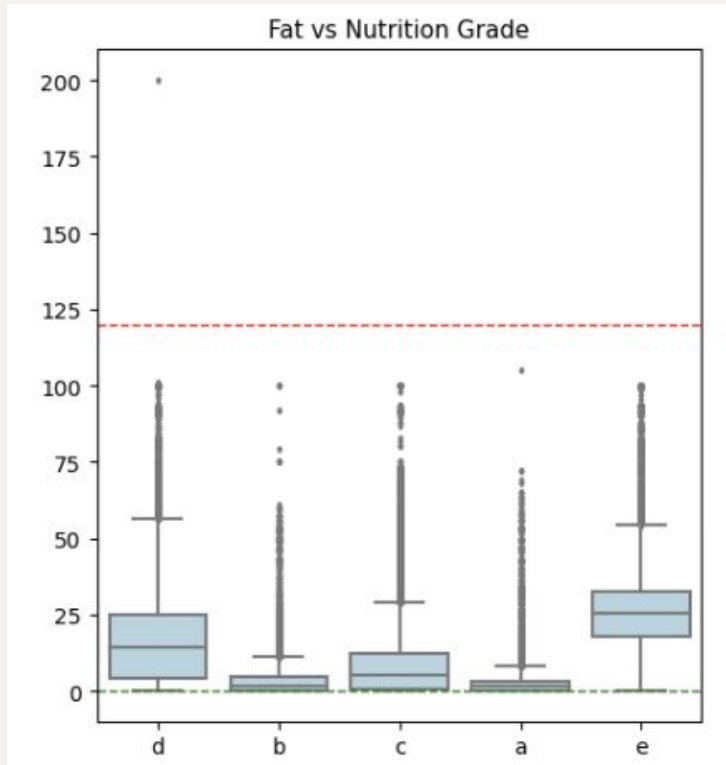
**Prepared features
for modeling
(encoding, ratios,
scaling)**

Handling Missing Data

- Dropped all columns with **>80%** missing values.
- Kept only core nutritional features + key categories needed for modeling.
- **Categorical columns**
 - filled with "Unknown" to avoid losing rows.
- **Numeric columns**
 - filled using median imputation.



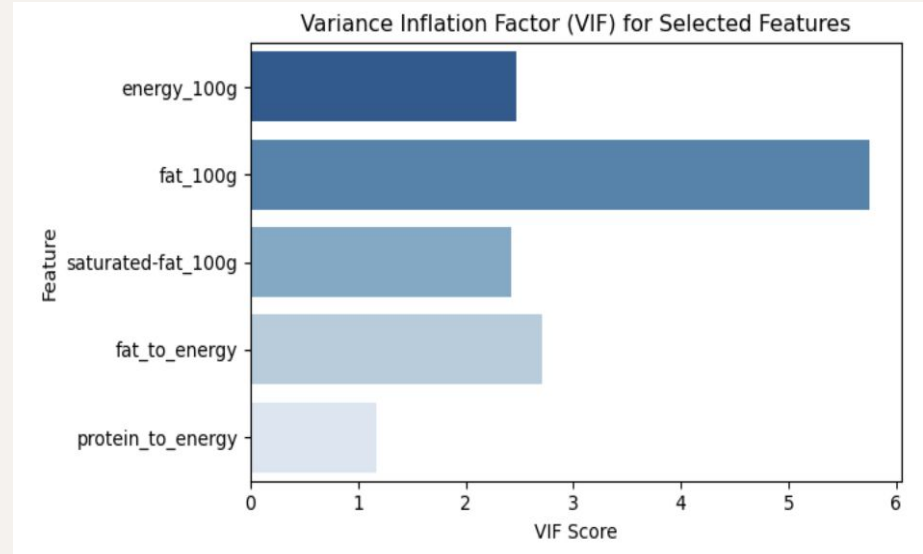
Outlier Treatment & Scaling



- Outlier capping for:
 - *energy_100g, fat_100g, sugars_100g, salt_100g, etc.*
- Applied **RobustScaler** to normalize nutrients while reducing the impact of extreme values
- Ensures stable modeling performance, especially for Logistic Regression

Feature Engineering & Encoding

- Created engineered features:
 - *fat_to_energy*,
 - *protein_to_energy*,
 - *sugar_to_carb*
- palm oil flags
- additives count
- One-hot, count, and label encoding
- Final dataset: fully numeric, consistent, and ready for modeling
 - Rows: 237836
 - Features: 41



Initial Models

Logistic Regression

=== Logistic Regression Results ===

	precision	recall	f1-score	support
1.0	0.631	0.736	0.679	9369
2.0	0.617	0.468	0.532	13543
3.0	0.500	0.475	0.487	9864
4.0	0.494	0.696	0.578	7215
5.0	0.735	0.662	0.696	7577
accuracy			0.588	47568
macro avg	0.595	0.607	0.594	47568
weighted avg	0.595	0.588	0.585	47568

```
[[6892 1740 164 559 14]
 [3712 6335 3119 276 101]
 [ 294 1932 4685 2261 692]
 [ 12 210 969 5023 1001]
 [ 11 52 441 2059 5014]]
```

Random Forest

=== Random Forest Results ===

	precision	recall	f1-score	support
1.0	0.974	0.955	0.964	9369
2.0	0.936	0.961	0.948	13543
3.0	0.906	0.915	0.911	9864
4.0	0.905	0.885	0.895	7215
5.0	0.962	0.948	0.955	7577
accuracy			0.937	47568
macro avg	0.937	0.933	0.935	47568
weighted avg	0.937	0.937	0.937	47568

=== Baseline Random Forest Performance ===

Training Accuracy: 0.9999

Test Accuracy: 0.9366

Training Macro F1: 0.9999

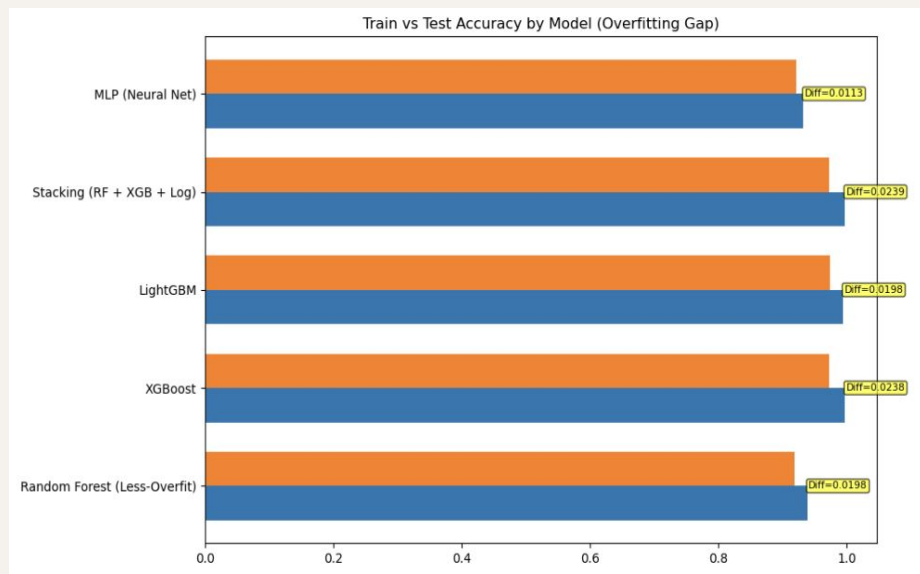
Test Macro F1: 0.9346

Train/Test Split + Models Evaluated

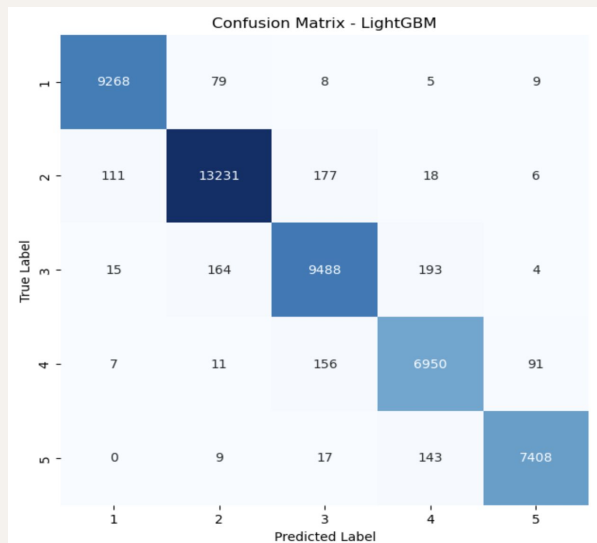
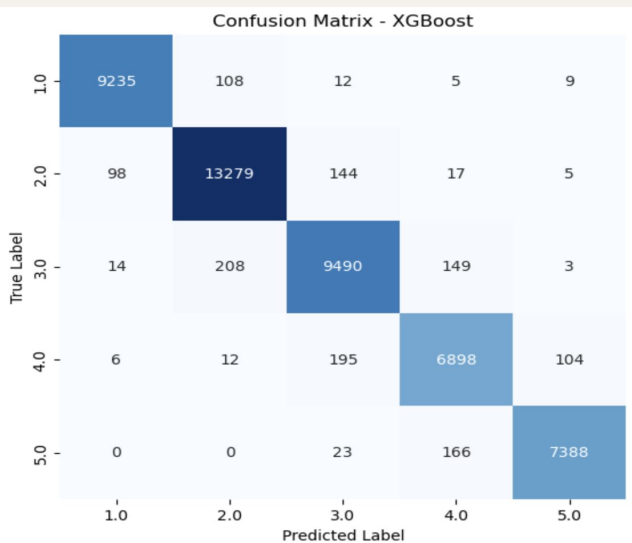
80/20 split, random_state=42 for reproducibility.

No validation set needed due to large dataset + cross-validation later.

Model	Training Accuracy	Testing Accuracy	Training Macro F1	Testing Macro F1
Logistic Regression	0.5891	0.5829	0.5946	0.5887
Random Forest	0.9383	0.9185	0.9348	0.9152
LightGBM	0.9941	0.9743	0.9940	0.9735
XGBoost	0.9969	0.9731	0.9967	0.9722
Stacking	0.9967	0.9728	0.9965	0.9719



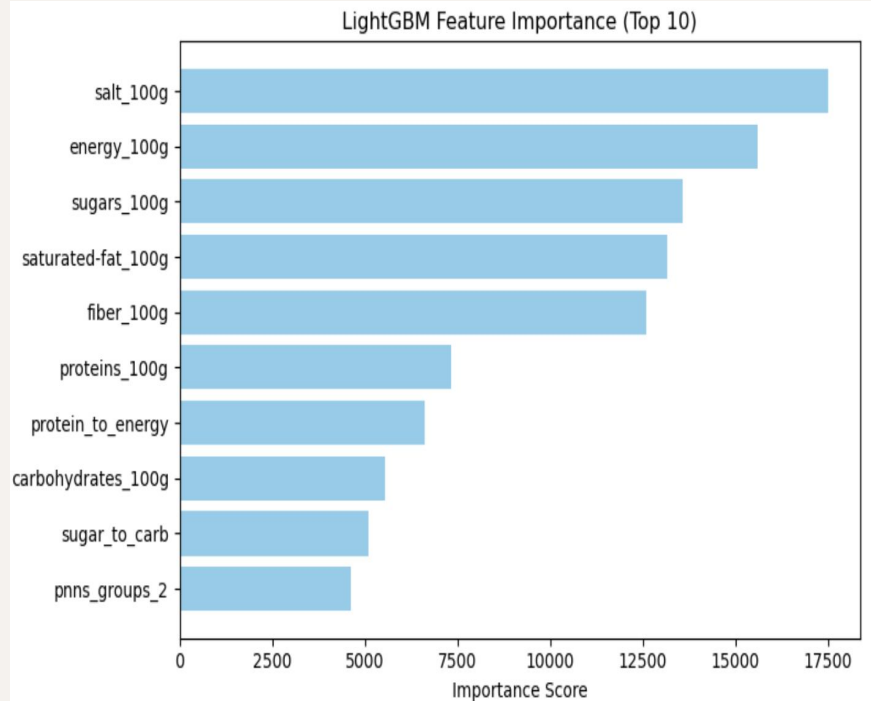
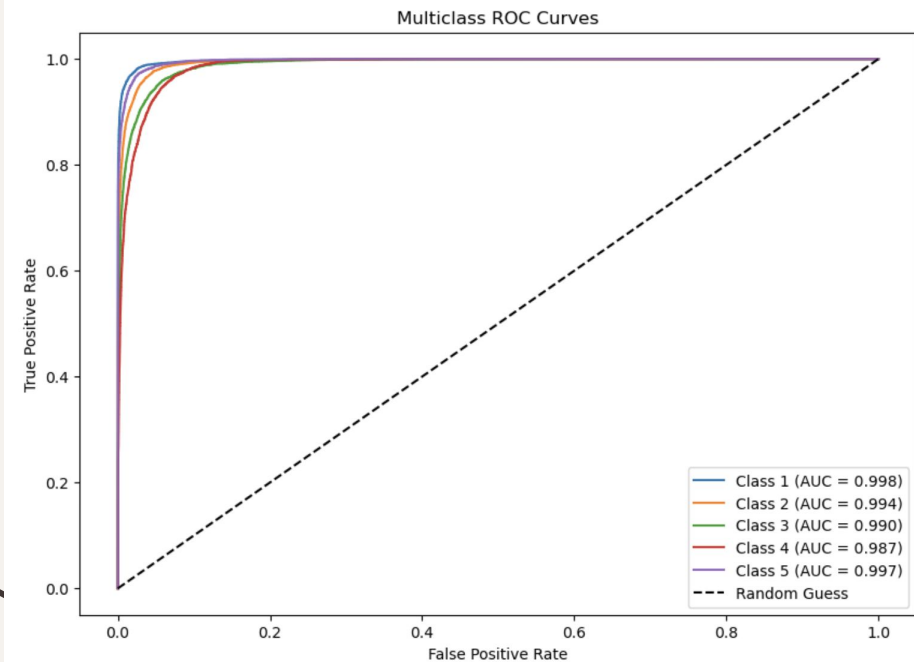
Model Performance Overview



- Both models show strong diagonals with high accuracy.
- Most mistakes happen only between neighboring classes.
- Extreme misclassifications are nearly zero.

ROC Curve/Feature Importance

Macro AUC: 0.9931



Interpretation, Limitations & Final Answer

Interpretation of Results

- Nutri-Score is highly predictable from nutrient values.
- Models captured the true scoring patterns very accurately.

Limitations

- OpenFoodFacts data contains noise.
- Single train/test split; hyperparameter tuning could be deeper.

Future Work

- Add 70/15/15 split, broader tuning, external validation set.

Final Answer

- *Yes, machine learning can reliably predict Nutri-Score labels.*
- LightGBM & XGBoost demonstrated excellent accuracy and consistency.

References

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